

ICU BIOMEDICAL COMPILER

LEX / YACC / SDT / 3AC / CSE / MCU

● CSE COMPLETE

LIVE PATIENT TELEMETRY

HR	SpO2	TEMP	SYS	DIA	GLUCOSE
112	130	38.7	145	92	126

HEALTH RISK EXPRESSION

(HR > 100) && (SpO2 < 92)

NORMAL TEST

CSE TEST

CORRUPTED INPUT

RANGE TEST

RESET

COMPILER PHASES - CLICK ANY STEP

1 LEX

2 YACC

3 SDT

4 SEMANTIC

5 3AC

6 CSE

7 MCU

RUN ALL

ANALYSIS / PARSE TREE / 3AC

BEFORE OPTIMIZATION

```
t1 = HR > 100
t2 = SpO2 < 92
t3 = t1 && t2
```

AFTER CSE

```
t1 = HR > 100
t2 = SpO2 < 92
```

OPTIMIZATION / TARGET / RESULTS

CYCLE COMPARISON

```
Before Optimization : 12 cycles
After Optimization  : 12 cycles
Cycle Reduction     : 0.0%
Instructions Removed: 0
```

WHY CSE?

Repeated calculations are reused instead of

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● 3AC COMPLETE

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ANALYSIS / PARSE TREE / 3AC

```
=====
THREE ADDRESS CODE / QUADRUPLES
=====
```

THREE ADDRESS CODE

```
01. t1 = HR > 100
02. t2 = SpO2 < 92
03. t3 = t1 && t2
```

OPTIMIZATION / TARGET / RESULTS

3AC PURPOSE

Three Address Code breaks a complex medical expression into simple instructions.

Example:

```
t1 = HR > 100
t2 = SpO2 < 92
t3 = t1 && t2
```

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• SEMANTIC COMPLETE

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COMPILER PHASES - CLICK ANY STEP

1 LEX	2 YACC	3 SDT	4 SEMANTIC	5 3AC	6 CSE	7 MCU	RUN ALL
-------	--------	-------	------------	-------	-------	-------	---------

ANALYSIS / PARSE TREE / 3AC

```
=====
SEMANTIC ANALYSIS
=====

OK - Identifier and type validation passed.

PATIENT RANGE CHECK
-----
! SpO2 = 130.0 is outside safe range 70 - 100 %
```

OPTIMIZATION / TARGET / RESULTS

SAFETY CHECK SUMMARY

```
-----
HR      112.0  SAFE
SpO2    130.0  OUT OF RANGE
TEMP    38.7   SAFE
SYS     145.0  SAFE
DIA     92.0   SAFE
GLUCOSE 126.0  SAFE
```

ICU BIOMEDICAL COMPILER

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• SDT COMPLETE

LIVE PATIENT TELEMETRY

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RANGE TEST

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COMPILER PHASES - CLICK ANY STEP

1 LEX	2 YACC	3 SDT	4 SEMANTIC	5 3AC	6 CSE	7 MCU	RUN ALL
-------	--------	-------	------------	-------	-------	-------	---------

ANALYSIS / PARSE TREE / 3AC

```
-----
NAME      TYPE    MIN    MAX    UNIT
-----
HR         INT     30     220   bpm
SpO2       INT     70     100   %
TEMP       FLOAT   30.0   43.0   C
SYS        INT     50     200   mmHg
DIA        INT     30     130   mmHg
GLUCOSE    FLOAT   40.0   400.0  mg/dL
```

OPTIMIZATION / TARGET / RESULTS

The parsed identifiers are connected to medical symbol-table attributes:

```
TYPE      : INT / FLOAT
MIN / MAX : physiological boundary
UNIT      : medical measurement unit
```

These attributes are used during semantic validation and intermediate-code generation.

ICU BIOMEDICAL COMPILER

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• YACC COMPLETE

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ANALYSIS / PARSE TREE / 3AC

```
=====
YACC-STYLE PARSER
=====

GRAMMAR IDEA
-----
expression -> AND { OR AND }
comparison -> arithmetic [comparison arithmetic]
arithmetic -> term { + or - term }
term        -> factor { * or / factor }
```

OPTIMIZATION / TARGET / RESULTS

```
SYNTAX TREE
-----
^-- &&
  |-- >
    |-- HR
    ^-- 100
  ^-- <
    |-- SpO2
    ^-- 92
```

ICU BIOMEDICAL COMPILER

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• LEX COMPLETE

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ANALYSIS / PARSE TREE / 3AC

TOKEN TYPE	TOKEN VALUE
LPAREN	(
ID	HR
GT	>
NUMBER	100
RPAREN	)
AND	&&
LPAREN	(
ID	SpO2

OPTIMIZATION / TARGET / RESULTS

```
LEX PURPOSE
-----
The scanner converts the medical expression
into tokens such as identifiers, numbers,
operators and parentheses.

Biomedical identifiers:
HR, SpO2, TEMP, SYS, DIA, GLUCOSE
```

ICU BIOMEDICAL COMPILER

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● MCU CODE READY

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ANALYSIS / PARSE TREE / 3AC

```
01. LOAD R1, HR
02. CMP_GT R1, 100
03. MOV t1, R1
04. LOAD R1, SpO2
05. CMP_LT R1, 92
06. MOV t2, R1
07. LOAD R1, t1
08. AND R1, t2
09. MOV t3, R1
--
```

OPTIMIZATION / TARGET / RESULTS

Architecture : Simplified MCU
Mode : LOW POWER
Registers : R1 + temporaries
Optimization : CSE

FINAL EXECUTION

Before : 12 cycles
After : 12 cycles